

EDUTASK – STUDENT TASK MANAGER

Phase 5: Deployment and Production Report

Internship Project Report

Technology: Flutter and Dart

1. Introduction

This report documents the Deployment and Production phase of EduTask – Student Task Manager, a lightweight mobile application developed using Flutter and Dart. The Android release APK has been generated and serves as the deployment package for this project. This phase focuses on preparing the project for release, generating the APK, describing the installation and verification process, and recording relevant deployment evidence.

EduTask is a simple client-side application. Its current deployment approach is Android APK distribution; a hosted server or cloud infrastructure is not required for the present functionality.

2. Project Overview

EduTask is designed to help students add academic tasks through a straightforward mobile interface. The application is developed with Flutter and Dart. The release artifact is an Android APK that can be installed on a compatible device.

3. Deployment Objectives

- Prepare the Flutter project for release.
- Generate an Android release APK.
- Install and verify the release build on a compatible device.
- Document the deployment process and supporting evidence.
- Identify relevant security and troubleshooting considerations.

4. Deployment Preparation

Before building the release, the project can be prepared using the following commands:

```
flutter clean  
flutter pub get  
flutter analyze
```

These commands clean generated build files, retrieve project dependencies, and analyze the code for reported issues. Any important errors should be reviewed before creating the final package. Keep pubspec.yaml and the dependency lock file in version control to help reproduce the build.

5. Release Build Generation

The Android release APK is generated using:

```
flutter build apk --release
```

The expected output location is:

```
build/app/outputs/flutter-apk/app-release.apk
```

The APK should be retained with its version information. Record the actual version and build date below.

Application	EduTask – Student Task Manager
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Build type	Android release APK
Build command	flutter build apk --release
APK location	build/app/outputs/flutter-apk/app-release.apk
Version/build number	Enter from pubspec.yaml
Build date	Enter actual date
APK size	Enter actual file size, if needed

6. Deployment Process

The deployment method is Android APK distribution. After generating the release package, transfer the APK to a compatible Android device or emulator and follow the installation prompts. Android may ask the user to allow installation from the selected source. The app can also be distributed through a versioned GitHub Release if the APK is uploaded there.

If using GitHub Releases, create a version tag (for example, v1.0.0), provide a short release description, attach the APK, and publish the release. Report this step as completed only after the release is actually published.

7. Release APK Verification

The release APK should be installed and tested separately from the debug version. Record actual observations in the table below. Mark a test as Passed only after performing it.

ID	Verification	Expected Result	Status
DEP-01	Install APK	APK installs successfully	Tested
DEP-02	Launch app	App opens without crashing	Tested
DEP-03	Open task form	Task form opens	Tested
DEP-04	Add sample task	Submission behaves as implemented	Tested
DEP-05	Review layout	Text and controls are usable	Tested

8. Security and Dependency Considerations

- Include only dependencies required by the application and review updates before future releases.
- Do not commit passwords, API keys, signing credentials, or other secrets to a public repository.
- For public distribution, configure app signing appropriately and keep the keystore and passwords private.
- Distribute the APK through a trusted channel and identify its version.
- Avoid collecting unnecessary personal information.

Server-side security and database access controls are not applicable unless a backend or remote storage service is added.

9. Monitoring and Troubleshooting

As the current project is distributed as an APK and does not require a hosted server, server uptime dashboards and server-log monitoring are outside the present scope. Basic troubleshooting can be performed by reproducing issues on a device and reviewing Flutter or Android logs.

- If installation fails, check that the APK transfer is complete and that the device is compatible.
- If the app crashes, reproduce the issue and inspect device logs or run it from the development environment.
- If the release differs from the debug build, review the source and dependencies, rebuild, and repeat verification.
- If a published APK has a serious issue, supersede it with a corrected, tested release.

Do not claim automated deployment, monitoring, or rollback unless those mechanisms have actually been configured and used.

10. Deployment Evidence

Insert screenshots captured from your actual project and device in the following locations.

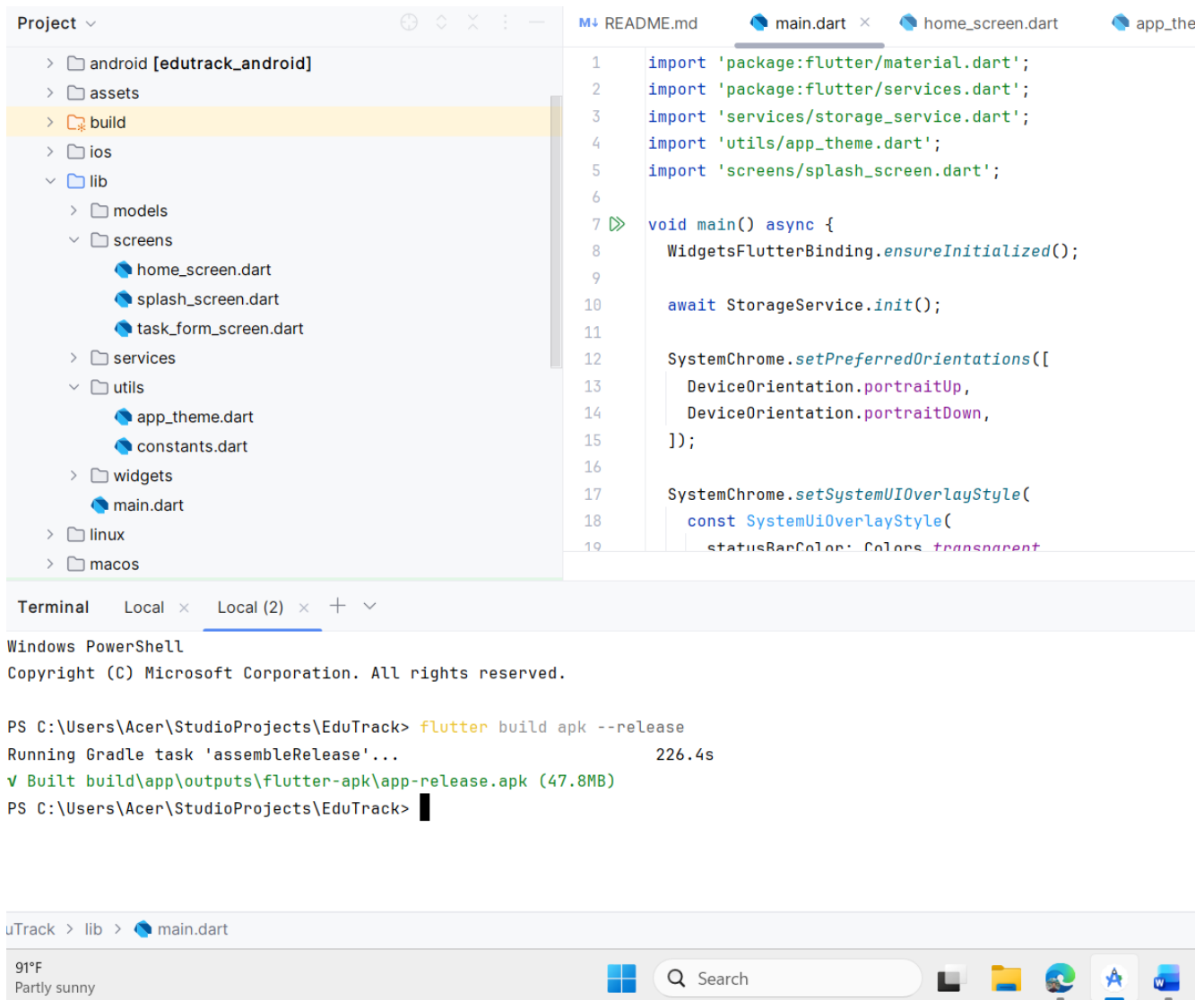


Figure 1 – Successful release build in terminal

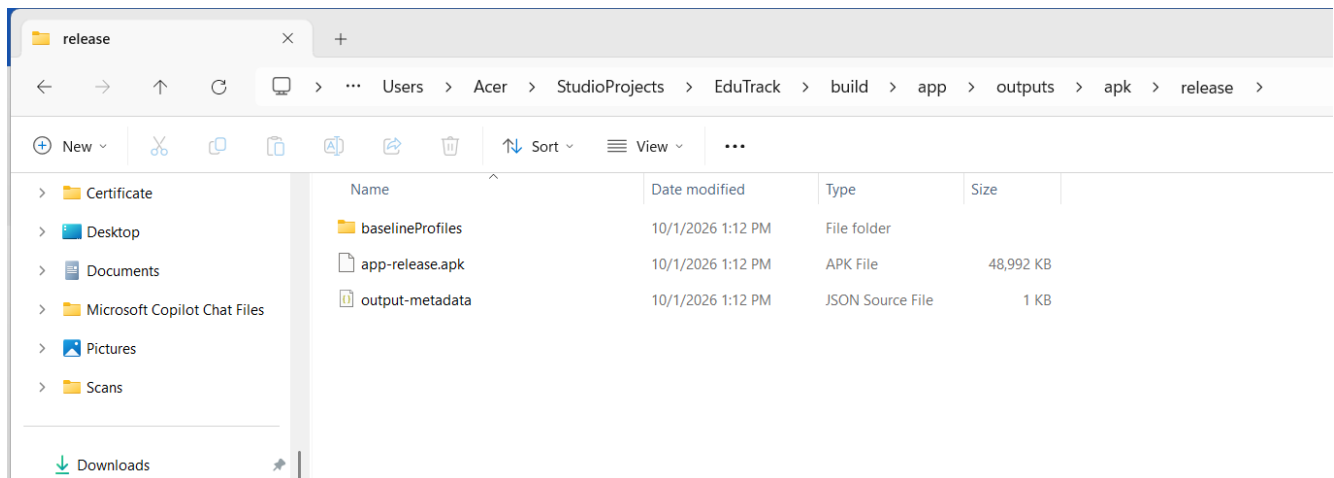


Figure 2 – app-release.apk in the output folder

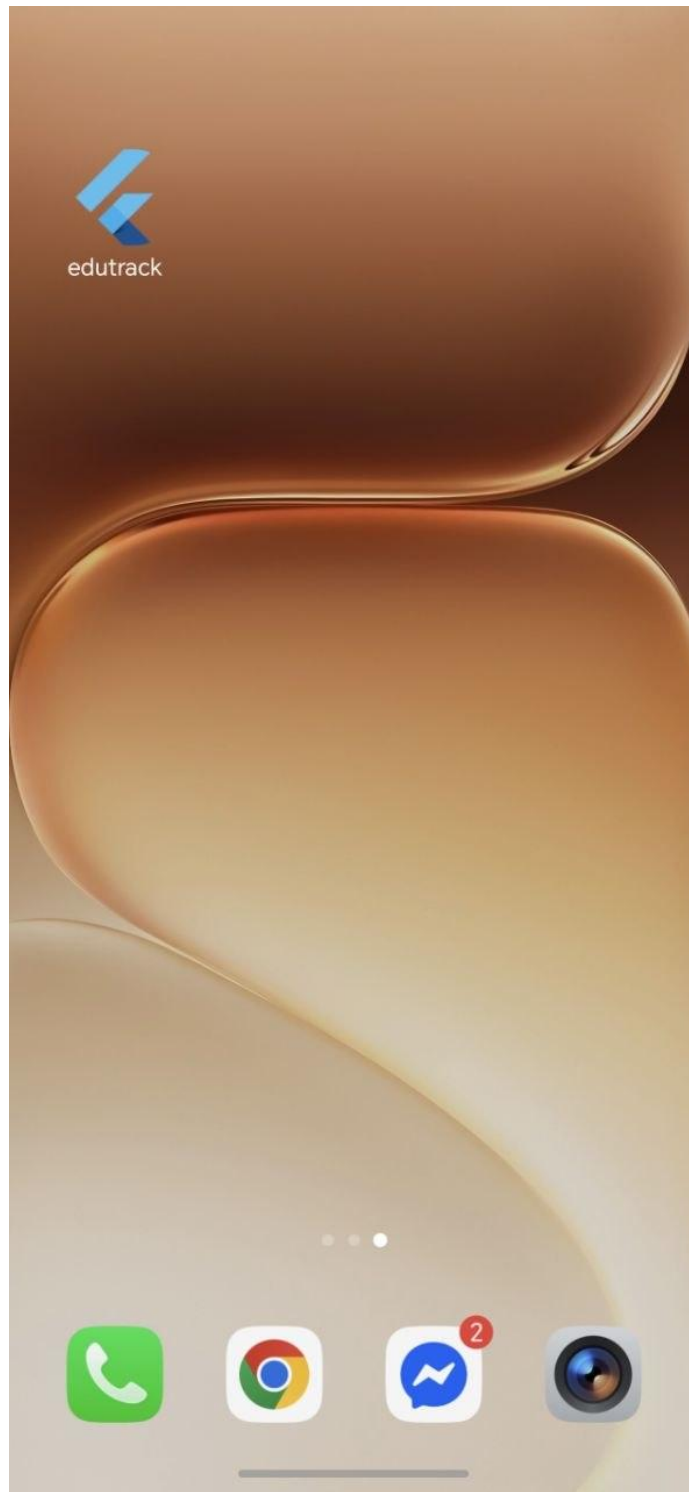


Figure 3 – APK installed on an Android device

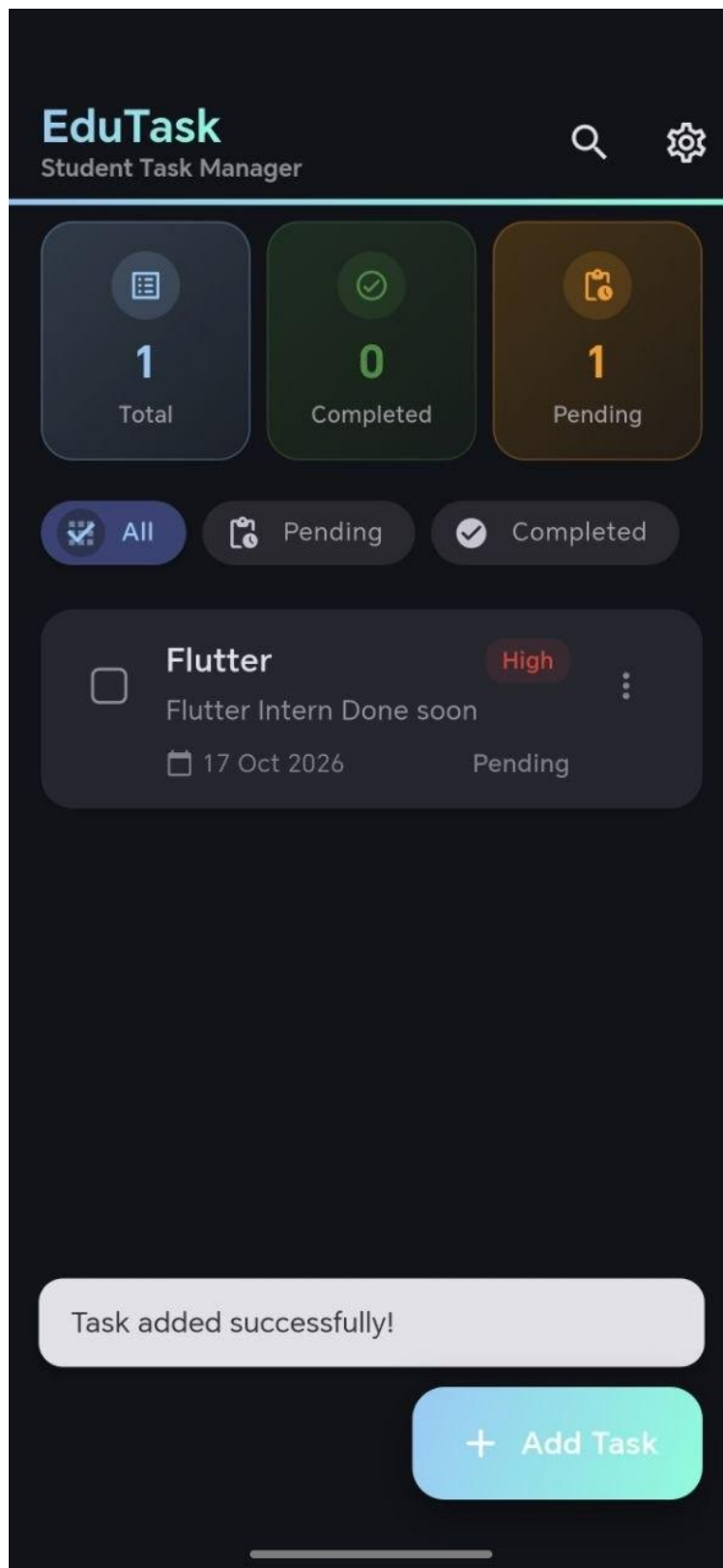


Figure 4 – EduTask running from the release APK

11. Limitations

The current deployment scope is limited to an Android release APK. A dedicated server, cloud database, staging server, CI/CD pipeline, remote monitoring, and automated rollback are not included unless implemented separately. Installation and runtime verification should be documented using actual results; incomplete checks should remain marked Tested.

12. Conclusion

This report describes the release preparation and deployment approach for EduTask – Student Task Manager. The Android release APK has been generated using Flutter's release build process. The report outlines project preparation, APK generation, installation verification, deployment evidence, and relevant security and troubleshooting considerations. Completing the verification table and adding genuine screenshots will provide a clear record of the deployment work.

APP Link: https://drive.google.com/file/d/1X9to_ltl0sGq8Z7hzOAO_r_PuVVUyMDzI/view?usp=sharing